

Three-Exam Prep Blueprint

Coal India MT (Systems) | IBPS SO IT Officer (Scale I) | DSSSB TGT Computer Science

Built from your uploaded official syllabus PDFs. Study order = shared-first; general sections kept separate from CS/IT. DSSSB = DOE (governing body).

Source (read first). This document is built from your uploaded official PDFs: Coal India *System Discipline* syllabus (GATE-CS style), the **IBPS CRP SPL-XVI** notification (Structure-of-Examination pages, dated 30.06.2026), and **two** DSSSB TGT CS syllabi (NDMC and DOE / pc_56-24).

✓ **IBPS pattern below is the current SPL-XVI cycle, confirmed from the notification itself** — Professional Knowledge is now tested in **both** Prelims (25Q/50M) and Mains (50Q/100M), and Mains adds a **Descriptive English paper** (2Q/25M). In the Main, all tests except Professional Knowledge are qualifying; merit is built on the PK score.

1 Study Order — CS/IT Core, Shared Topics First

This ladder covers the **CS/IT technical core only**, ranked by how many of the three exams test each topic. The **general sections** (English, Reasoning/Aptitude, Numerical/Mathematics, Hindi, Pedagogy) are kept **separate in Section 1B** below, so your domain prep and your aptitude prep never get tangled.

Tier	Topics	Appears in	Why this order
TIER 1 Shared core	Programming (C/C++) & Data Structures • DBMS & SQL • Operating Systems • Computer Networks • Computer Organization & Architecture • Digital Logic / Electronics • Algorithms • Software Engineering	Coal India IBPS SO DSSSB (DOE)	Single biggest ROI — one prep serves all three. Start here.
TIER 2 Two of three	Web tech / HTML / CSS / XML • Cyber Security & Computer Network Security • basics of Cloud / AI-ML	IBPS + DSSSB (DOE)	Strong secondary payoff; slot after the shared core is solid.
TIER 3 Exam-specific (CS/IT)	Coal India only: Theory of Computation, Compiler Design (deep, GATE-level). IBPS only: deeper Cloud Computing & Cyber Security. DSSSB (DOE) only: Java & website design, .NET, Front-end design tools, OOP/Numerical Techniques, Computer Graphics & Multimedia, MIS, E-Commerce, Mobile Computing, Linux, Internet/Web programming.	One exam each	Highest effort-per-mark; layer in per your target exam & date.

Your self-assessment: Computer Networks and Theory of Computation / Compiler are “lost”; DBMS and OS need revision. CN sits in Tier 1 (rebuild first, biggest lever). TOC/Compiler is Tier 3 / Coal-India-only — deep but defer until the shared core is done.

1B General Sections — Kept Separate From CS/IT

These are the non-technical scoring sections. They share a common aptitude spine across all three exams, but each exam adds its own extras. Prep these in a **separate daily slot** from your CS core.

General section	Coal India (Paper I)	IBPS SO IT (SPL-XVI)	DSSSB TGT (DOE)
English Language	✓ General English	✓ Prelims + Mains + Descriptive paper	✓ English Lang. & Comprehension; Foundation Course in English
Reasoning / General Intelligence	✓	✓ Prelims + Mains	✓ General Intelligence & Reasoning
Numerical / Quantitative / Mathematics	✓ Numerical Ability	✓ Quant, Prelims + Mains	✓ Arithmetical Ability + Mathematics I-IV , Statistical Techniques, Interpolation
General Awareness / GK	✓ GK, coal sector, SDGs, computer awareness	— (Quant instead for IT)	✓ General Awareness
Hindi Language	—	—	✓ (as applicable to DSSSB)

General section	Coal India (Paper I)	IBPS SO IT (SPL-XVI)	DSSSB TGT (DOE)
Business / Commerce papers	—	—	✓ Business Communication, Business Organization & Management, Business Economics, Financial Accounting, Knowledge Management
Teaching Methodology / Pedagogy	—	—	✓ (DSSSB teacher-post component)

Note: **Mathematics I-IV** (DSSSB DOE) is a heavy, degree-level maths block — it lives here in general/quantitative, not in the CS core. It partly overlaps Coal India's Engineering Mathematics, so revising it helps both.

2 IT / CS Topic Map — Similarities & Differences

✓ = tested • depth noted where it differs. Coal India = GATE-level; IBPS = graduation-level objective; **DSSSB = DOE syllabus (governing body)**. This map is the **CS/IT core** only — general sections are in Section 1B.

CS/IT Subject	Coal India (Systems)	IBPS SO IT (SPL-XVI)	DSSSB TGT (DOE)
Programming / Data Structures	deep	✓	C, C++, Java
DBMS & SQL	deep	✓	✓
Operating Systems	deep	✓	✓
Computer Networks	deep	✓	+ TCP/Protocols
Computer Org. / Architecture	✓	some	✓
Digital Logic / Electronics	✓	misc	✓
Algorithms / DAA	✓	✓	✓
Software Engineering	some	✓	✓
OOP / Numerical Techniques	via prog	OOP	✓
Theory of Computation	✓	—	—
Compiler Design	✓	—	—
Cyber Security / Network Security	some	✓	✓
Cloud / AI-ML	—	✓	MIS/e-com
Web: HTML/CSS/XML	—	web tech	✓
Java web / .NET / Front-end tools	—	—	✓
Computer Graphics / Multimedia	—	—	✓
MIS / E-Commerce / Mobile Computing	—	—	✓
Linux Environment	—	OS	✓

✓ full / deep **partial** lighter or sub-topic only — not in syllabus. This is the **CS/IT core only**; general sections are in Section 1B.

Key takeaways

- **DSSSB governing body = DOE**, so the DOE syllabus is authoritative here; NDMC is used only as a more detailed reference for explaining topics (Section 3C).
- The DOE TGT CS syllabus is **degree-level** and overlaps heavily with Coal India + IBPS — the shared core serves all three.
- **TOC + Compiler** are unique to Coal India.
- DSSSB-only CS extras (Java/.NET/web, graphics, MIS, e-commerce, mobile computing) are Tier-3.
- English, Reasoning, Numerical/Maths, GK, Hindi and Pedagogy are handled separately in Section 1B.

3 Exact Syllabus — As Per Your Uploaded PDFs

A. Coal India MT — System Discipline (Paper II)

Section	Topics
1. Engineering Mathematics	Discrete Math (logic, sets, relations, lattices, groups, graphs, combinatorics); Linear Algebra; Calculus; Probability & Statistics; Numerical Methods.
2. Digital Logic	Boolean algebra; combinational & sequential circuits; minimization; number representation & computer arithmetic.
3. Computer Org. & Architecture	Machine instructions & addressing; ALU, datapath, control unit; pipelining & hazards; memory hierarchy; I/O (interrupt, DMA).
4. Programming & Data Structures	C; recursion; arrays, stacks, queues, linked lists, trees, BST, heaps, graphs.
5. Algorithms	Searching, sorting, hashing; asymptotic complexity; greedy, DP, divide-and-conquer; graph traversal, MST, shortest paths.
6. Theory of Computation	Regular expressions & finite automata; CFG & PDA; regular/CF languages, pumping lemma; Turing machines, undecidability.
7. Compiler Design	Lexical analysis, parsing, syntax-directed translation; runtime environments; intermediate code; data-flow analysis.
8. Operating System	System calls, processes, threads, IPC, concurrency & synchronization; deadlock; CPU/IO scheduling; memory & virtual memory; file systems.
9. Databases	ER model; relational algebra, tuple calculus, SQL; integrity constraints, normal forms; file organization, indexing (B/B+ trees); transactions & concurrency.
10. Computer Networks	OSI & TCP/IP layering; switching; data link (framing, error detection, MAC, Ethernet); routing (shortest path, DV, LS); IPv4/CIDR, ARP/DHCP/ICMP, NAT; transport (flow/congestion, UDP, TCP, sockets); application (DNS, SMTP, HTTP, FTP).

Note on Paper I (common aptitude): General Knowledge/Awareness, Reasoning, Numerical Ability, General English (per CIL notification). The System PDF above is Paper II only and is stated to be “indicative, not exhaustive.”

B. DSSSB TGT Computer Science — DOE syllabus (governing body, authoritative)

Source: pc_56-24 (DOE). 32 heads, degree-level. Below they are split into **CS/IT core** and **general** so the two prep tracks stay separate.

CS / IT core heads

Computer Basics & PC Software • Fundamentals of IT • Programming with C, Data Structures using C, C++ • Object/Computer Oriented Programming / Numerical Techniques • Problem Solving & Programming • Digital Electronics • Computer Architecture • Operating Systems • DBMS • Computer Networks • TCP / Protocols • Design & Analysis of Algorithms • Software Engineering • Java Programming & Website Design • .NET Programming • Front-End Design Tools • Internet Programming (Web Server, CSS, XML/XHTML, DTD) • Linux Environment • Computer Network Security • Computer Graphics & Multimedia • Management Information System • Mobile Computing • E-Commerce • Knowledge Management & New Economy.

General / non-CS heads (prep in your separate slot — see Section 1B)

Mathematics I, II, III, IV • Statistical Techniques (descriptive statistics, probability, distributions, inference) • Interpolation • Basics of Physics • Foundation Course in English • Business Communication • Business Organization & Management • Business Economics • Financial Accounting.

C. DSSSB TGT Computer Science — NDMC version (reference only)

Not your governing body. Included because it spells out topics in more granular, CBSE-style detail — useful as a **study reference** to understand what each DOE head contains (e.g. exactly which OS or CN sub-topics), plus it shows the pedagogy block DSSSB teacher posts carry.

Head	Topics
Computer Systems & Organization	Hardware/software, CPU, memory units; system vs application software; OS functions & UI; Boolean logic (NOT/AND/OR/NAND/NOR/XOR, De Morgan).
Systems Architecture	System buses; internal memory RAM/ROM; I/O devices, I/O modules, programmed I/O, interrupt-driven I/O, DMA, channels, external interface.
Operating Systems	OS organization, device/process management, scheduling, synchronization, deadlocks, memory & file management.
Computational Thinking & Programming (Python)	Problem solving, algorithms/flowcharts/pseudocode; Python basics, tokens, data types; functions, files (text/binary/CSV); dictionaries.
Computer Networks	Evolution (ARPANET/NSFNET); data communication terms; transmission media; network devices; LAN/MAN/WAN; topologies; Internet/URL/WWW; email, chat, VoIP.
Database Management & SQL	DB concepts; relational model & keys; SQL DDL/DML; data types & constraints; MySQL; Math/Text/Date functions.
Data Handling using Pandas	Series & DataFrames; creation, indexing, slicing; row/column operations; Matplotlib intro.
HTML & Web	HTML/DHTML, tags, links, tables, forms; DOM; CSS margins; XML (DTD).
Emerging Trends	AI, ML, NLP, AR/VR, Robotics, Big Data, IoT, sensors, smart cities.
Society, Law & Ethics / Societal Impacts	Digital footprint & etiquette; IPR, copyright, licensing (Creative Commons/GPL/Apache); cyber-crime & safety; IT Act; e-waste; health.
Teaching Education & Methodology (Pedagogy)	Learning & Teaching; Language across the curriculum; Understanding discipline & subject; Gender/school/society; Pedagogy of a school subject; Knowledge & curriculum; Assessment for learning; Inclusive school; Childhood & growing up; Drama & Art in education.

4 Priority Plan — Shared Core First

Ordered by your ranking (nearest exam → overlap efficiency) and your weak areas (CN + TOC/Compiler lost; DBMS + OS need revision).

Phase	Focus (CS/IT core)	Serves
Phase 0 Rebuild core	Computer Networks from scratch (your biggest “lost” lever, Tier-1) → then DBMS & OS revision → DS/Programming refresh. Add Digital Logic + COA.	All three
Phase 1 Deepen	Algorithms / DAA, Software Engineering, OOP. (Aptitude runs in parallel on the general track — see below.)	All three
Phase 2 Exam-specific	Coal India: Theory of Computation + Compiler (deep). IBPS: Cyber Security, Cloud.	CIL / IBPS
Phase 3 DSSSB (DOE) layer	DSSSB-only CS heads: Java & website design, .NET, front-end tools, Internet/Web programming, Computer Graphics & Multimedia, MIS, E-Commerce, Mobile Computing, Linux, Network Security.	DSSSB

Parallel general track (separate daily slot)

Run alongside from day one, independent of the CS phases: **Reasoning + Quantitative + English** daily (IBPS Prelims & Mains, CIL Paper I, DSSSB general). Add **IBPS Descriptive English** practice before Mains. Add **DSSSB extras** — Mathematics I-IV & Statistics, GK, Hindi, Business/Commerce papers, and Pedagogy — as its exam date firms up.

The efficiency logic: Phases 0-1 (the shared Tier-1/2 core) are ~70-80% of Coal India Paper II *and* the IBPS PK section *and* the bulk of DSSSB-DOE's technical heads. You build it once. Only Phase 2-3 split by exam.

5 IBPS SO IT Officer — Exam Pattern (SPL-XVI, confirmed)

This is the part that felt confusing — because the older SPL-XV pattern was different. Below is the **current SPL-XVI** structure, taken directly from your uploaded notification's Structure-of-Examination pages.

Three stages: **Preliminary** → **Main** → **Interview**. All objective tests carry **0.25 negative marking**; blanks are not penalised. Prelims is a qualifying gate. In the Main, **all tests except Professional Knowledge are qualifying** — **merit is built on the PK score**.

Prelims — IT Officer (Scale I) • 100 Q / 125 marks / 80 min

Test	Qs	Marks	Time
English Language	25	25	20 min
Reasoning	25	25	20 min
Quantitative Aptitude	25	25	20 min
Professional Knowledge	25	50	20 min
Total	100	125	80 min

→ **Professional Knowledge is now in Prelims** (25Q/50M — double weight per question). Each section has its own cut-off. Your CS rebuild pays off from the first hurdle, not just Mains.

Mains — IT Officer (Scale I) • 225 marks / 155 min

Test	Qs	Marks	Time
English Language	30	30	25 min
Reasoning	40	40	35 min
Quantitative Aptitude	30	30	25 min
Professional Knowledge (objective)	50	100	40 min
Descriptive Paper (English)	2	25	30 min
Total	150 + 2	225	155 min

→ PK (100M) is the **merit-deciding** section; English/Reasoning/Quant are qualifying only. The **Descriptive English paper (2Q/25M/30min)** is the one genuinely new prep item for you.

Interview & final merit

Only PK-driven Mains performance builds the merit rank for shortlisting; the final list combines Main + Interview in an **80:20** ratio (per the Interview clause). Eligibility: 4-yr Engg degree in CS/IT/ECE/EEE etc. (or relevant PG / DOEACC B-level), age 20–30 as on 01.07.2026. Fee ■175 (SC/ST/PwBD) / ■850 (others). One post only.

✓ **Confirmed from your SPL-XVI notification.** Prelims 100Q/125M/80min (incl. PK 25Q/50M); Mains 225M/155min (objective 150Q/200M + descriptive 2Q/25M); 0.25 negative marking; merit on Professional Knowledge. Prelims tentatively 29 Aug 2026, Mains ~1 Nov 2026 — confirm final dates on the admit card at ibps.in.

Verify against official portals: coalindia.in (Advt. 03/2026, System discipline) • ibps.in (SPL-XVI, admit-card dates) • dsssb.delhi.gov.in (confirm DOE vs NDMC post code). Built from your uploaded official PDFs, July 2026 — IBPS pattern confirmed from the SPL-XVI notification. Where anything differs from the live notification, the notification prevails.